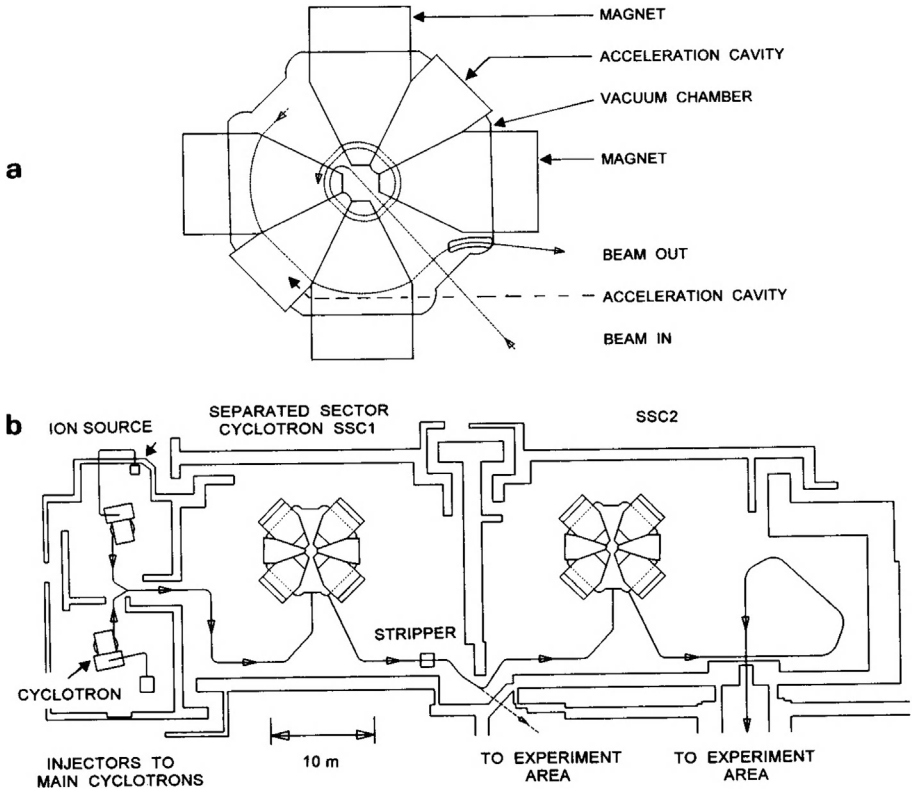


Figure 16.9 (a) Principle of a sector-separated cyclotron and (b) the GANIL accelerator complex (beam-line bending magnets are not shown).

16.7. FREQUENCY MODULATED CYCLOTRONS AND SYNCHROTRONS

The frequency of a cyclotron can be modulated to take into account the variation in velocity and mass as relativistic effects increase in importance. At very high energies the radius becomes very large. This has led to two somewhat different accelerator designs. The *frequency modulated* (FM) or *synchrocyclotron* maintains the original cyclotron principle with a spiral particle path, while in *synchrotrons* the particle path is fixed in a circular orbit; Figure 16.10 is an example of the latter.

In a synchrotron there is a balance between frequency and the magnetic field, one or both being varied during the acceleration (cf. (16.7a)). From (16.6) and (16.7) one obtains

$$E_{\text{kin}} = \nu\pi r^2 Bq \tag{16.12}$$

by eliminating the mass from the relation for the kinetic energy of the projectile. In these machines the particles are accelerated in bursts, since the frequency of the accelerating

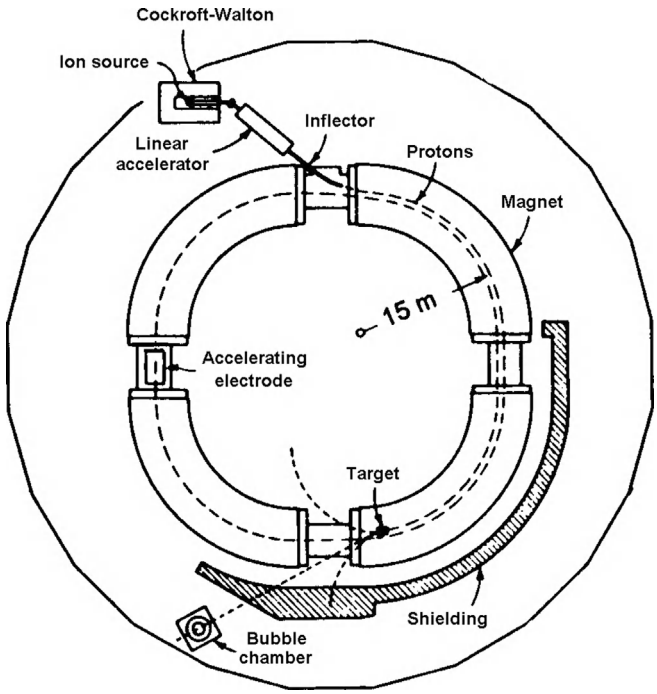


Figure 16.10 Principle of a modulated multistage circular accelerator.

potential must be modulated throughout the traversal of one beam burst in the machine, and only the particles in that burst would be in resonance with the frequency change of the accelerating gap. As a result, average beam currents are much smaller in FM cyclotrons and synchrotrons than in constant frequency cyclotrons.

Synchrotrons which accelerate positive particles use smaller accelerators, often linear, for injecting the beam into the circular synchrotron. Figure 16.10 shows a typical synchrotron design, although it specifically refers to the late Bevatron in Berkeley which provided acceleration of protons to 6.5 GeV (the mass increase of a 6.5 GeV proton is 6.9 u) and heavier ions like carbon or neon to about 2 GeV u^{-1} it illustrates the design of such machines. The Bevatron injectors were a 0.5 MeV Cockcroft-Walton machine followed by a 10 MeV multistage linear accelerator or the Super-HILAC heavy ion linear accelerator. The latter combination was called Bevalac. In the Bevatron, the potential difference over the accelerating step was only 1500 V. As the particles accelerate, the magnetic field continues to increase in such a way that the radius of the particle orbit remains constant. The frequency of the oscillating voltage applied to the accelerating electrode must also increase, since the time required for a complete circuit becomes shorter and shorter until the velocity of the particles approaches the velocity or light, after which the circuit time remains constant. The whole accelerating cycle requires several

seconds, during which time the particles make many millions of orbits. Each burst of particles may contain $\sim 10^{12}$ particles, and in the Berkeley machine there was one burst every 6 s. The successful operation of the synchrotron depends on the existence of phase stability.

The Bevatron (later the Bevalac) operated from 1954 to 1993. Larger and more recent (proton) accelerators are the 76 GeV machine of Serpukhov (USSR), and the 500 GeV machine at Batavia in the USA. The Batavia machine contains 1000 magnets along its 6.3 km orbit, each magnet weighing 10 tons.

At CERN, a machine was built that could produce 10^{13} 400 GeV protons s^{-1}; the acceleration diameter was 2.2 km. The injection time (with 10 GeV protons) was 23 s (to fill the acceleration ring), while the acceleration time was 9 s; for each cycle 2.5 MeV was added in energy. The field strength of the 200 bending magnets was 1.8 T and the power consumption 50 MW.

Still higher energies have been achieved in collisions between particles circulating in two beam trains in opposite direction by using intersecting storage rings (ISR). In this system the energy loss due to momentum conservation in the reaction between high speed projectiles and stationary target atoms is eliminated. At CERN proton collision energies of 7 TeV have been obtained and the system is being upgraded to 8 TeV in 2012.

Synchrotrons have been built in which electrons are accelerated in a fixed orbit in a strong magnetic field ($\geq$1 T) to energies in the GeV range. The intense electromagnetic (synchrotron) radiation has usually a maximum intensity at $\leq$1 keV and is used for studies of photon induced reactions (e.g. for meson production), for radiation chemistry, as a strong light source, etc.

A special type of synchrotron is the so called storage ring. In a storage ring the particles are kept at a constant velocity generating a constant flux of syncrotron radiation. In order to maximise the synchrotron radiation intensity these rings are often equipped with wrigglers or undulators which give the beam multiple direction changes, see §16.9.1.

16.8. NEUTRON GENERATORS

Many radioactive isotopes are most conveniently produced by irradiation of a target with neutrons. There are three main ways of obtaining useful neutron fluxes:

(a) through accelerator-induced reactions (Table 16.1);
(b) through nuclear reactors (Ch. 19);
(c) through nuclear reactions induced by radioactive isotopes (§10.9).

16.8.1. Reactions

Table 16.1 gives the most common accelerator-induced nuclear reactions leading to the production of neutrons. In general, the neutron yield increases with increasing beam energy up to some maximum value of the latter after which competition of other nuclear